Validation and Reproducibility Report

Submitted with: Reducing Hallucinations in Large Language Models Through a Neuro-Symbolic Reasoning Layer

Pravin Anbalagan and Athiban Karthik, Vellore Institute of Technology

Generated 28 October 2026, 10:23 UTC from repository commit 8138fd2. Paper: 6 pages, IEEE A4 conference format.

1. Purpose

This report accompanies the paper and states what was verified, how, and what remains open. It exists so a reviewer can check the paper's numbers without re-running the experiments, and so the claims the evidence does not support are visible in one place rather than inferred from omissions.

Three principles govern it. Every figure in the paper is recomputed from a stored per-question or per-claim record, not transcribed. Where a result does not generalize, the paper says so. Where a mechanism is unimplemented or unmeasured, it appears in section 7 below rather than being left out.

2. What the paper claims, and what it does not

Claimed. On a controlled benchmark of generated enterprise worlds, replacing model inference with symbolic inference over the same retrieved evidence reduces unsupported assertions without buying the reduction through abstention. The reduction is measured against the same corpus and questions answered by a 7B instruction-tuned model.

Not claimed. General-domain factuality. Elimination of hallucination. Transfer to domains whose rules cannot be written down. Effectiveness across models, datasets or domains not tested here. The worlds are synthetic, with gold answers from an independent oracle, and are not real customer traffic.

Explicitly reported as not transferring. The semantic-parser configuration's selective accuracy of 1.000 on the frozen test split falls to 0.909 on the confirmatory split. The paper reports the parser's advantage as demonstrated on one split and not established on unseen worlds.

Reported as near-tautological. The gated system in Study II refutes none of its released claims. That holds by construction - the gate releases only what the graph confirms - and the paper says so in the abstract rather than presenting it as a finding.

3. Reproducibility

Decoding is greedy and every system in a run uses the same model, corpus and authorization context, so a run repeats exactly on one machine. The commands below regenerate every number, table and figure in the paper from the stored records.

```
python scripts/audit_metrics.py --tag paper --split test
python scripts/independent_metrics.py
python scripts/verify_claims.py
python scripts/make_audit_reports.py
python scripts/make_submission_report.py
python scripts/build_paper_docx.py \
    --template docs/paper/templates/ieee_a4_transitional.docx
```

Check	Scope	Result

Published metrics recomputed from stored records	55 values	**0 mismatched**
Metrics recomputed independently of the harness, from the definitions as written	151 values	**0 mismatched**
Experimental claims checked against code, tests and records	22 claims	**all substantiated**
Paired McNemar tests recomputed from per-question predictions	17 comparisons	**0 disagreed**
Database-free test suite	128 tests	**all passed**
Architecture figure type size at printed scale	min 6.0 pt	**at or above the 6 pt floor**

The remaining tests require a PostgreSQL instance and are excluded here rather than reported as passing unrun. The symbolic reasoner, the verification layer, the ablation-reach guards, the football verifier and the from-scratch language model are all covered by the database-free subset.

4. Statistical validation

Paired comparisons use the exact two-sided McNemar test. Cases where both systems agree carry no information about which is better, so of the discordant pairs only, the count favouring one system is tested against Binomial(n, 1/2). The exact binomial tail is used rather than the chi-square approximation, because the discordant counts are small.

All **17 comparisons** were recomputed from the per-question predictions rather than read from the stored value; **0 disagreed**. The two quoted in the paper were additionally checked by hand against the closed form:

Comparison	Discordant split	Closed form	p
Test split, parser vs embedding	40 against 2	2(1+42+861) / 2^42	4.11e-10
Confirmatory split, parser vs embedding	7 against 1	2(C(8,0)+C(8,1)) / 2^8 = 18/256	0.0703

At the 0.05 level the first is persuasive and the second is not. The paper states that distinction and does not claim the parser's advantage generalizes. No third-party implementation was available as a cross-check in this environment, so the closed-form arithmetic above serves instead.

One coincidence was checked rather than assumed: the confirmatory split's comparison against retrieval gives 168 against 16, **identical to the test split**. The two splits share no case identifiers and use different worlds, and both figures reconcile with their own accuracies. It is a coincidence.

Full table of all comparisons: docs/audit/STATISTICAL_VALIDATION.md

5. Verification-gate evidence

The ablation table shows the verification gate changing no decision-level metric, which invites the reading that it does nothing. Two different measurements are at stake. The ablation rows grade the structured decision, which the symbolic layer fixes before any model writes a word; the gate acts on prose. A separate controlled experiment measures the gate at the level of released text.

Property	Measured
Ablation switch reaches every guard layer	3 reads across 2 modules
Cases in both arms, identical ids, worlds and categories	255
Released answer differs between arms	**52**
Answers carrying a figure the evidence does not support	**4 gated, 6 ablated**
Decision-level fields differing between arms (8 fields x 255 cases)	**0**
Systems differ only by the ablation flag	frozenset({'no_verification'})
Ablation-reach regression tests	**9 passed**

The guard is layered by design, and reaching it took three attempts: the realizer refuses wording that introduces an unsupported figure, the controller re-verifies what the realizer released, and the release check withholds an answer whose report failed. The switch originally reached only the third, and two successive runs produced byte-identical arms because each remaining layer repaired what the ablated one let through. A regression test now asserts the switch reaches all three; a test checking only the default would have passed throughout the period the defect existed.

Unchanged decision metrics do not mean the gate has no effect. They mean this benchmark grades decisions, not prose. Quantifying the gate's contribution further requires a prose-faithfulness metric, which is listed as unresolved.

Full evidence: docs/audit/GATE_EVIDENCE.md

6. Corrections made during audit

Each entry states what was wrong and what evidence settled it. Nothing was changed on suspicion alone.

6.1 The accuracy numerator

Section VI reported accuracy as 241/255 = 0.945 and then explained it with counts that did not reach 241. The explanation accounted for the answerable cases and the not-covered ones and stopped: **205 + 6 is 211, not 241.** The thirty conditional and denied cases that the numerator includes were named in the breakdown of N but never carried into the sum.

Recomputed from the 255 per-question records for that configuration, every term checks and the numerator is right - it was the explanation that was incomplete:

Gold kind	n	Correct	Correctness rule in the scorer
Answerable (decision, value, result)	210	**205**	Answered and matching the gold value. All 205 definite answers were correct, hence selective accuracy 1.000.
Conditional	15	**15**	Outcome conditional or clarification, and not definite - a hedged answer naming the missing fact.
Denied	15	**15**	Outcome denied and no disclosure. Disclosure rate is measured on exactly this subset.
Not-covered	15	**6**	Outcome insufficient. Nine of fifteen answered instead: the failure reported in Section X.

205 + 15 + 15 + 6 = 241, which equals the 241 records independently marked correct, giving 241/255 = 0.945. Each group has a distinct correctness rule in the scorer, so none is a free pass: a conditional case fails if the system commits definitely, and a denied case fails if it leaks. Section VI now prints all four terms and the sum, and `scripts/verify_claims.py` fails the build if they do not close.

6.2 Scientific consistency

#	Correction	Evidence
1	**Unverifiable claims are no longer counted as false.** The reported rate was 0.505; the refuted rate is **0.467**, with a further **0.038** unverifiable.	The stored report counts every non-supported claim as false, folding 40 unverifiable claims in with 493 refuted ones. The paper states the opposite - a claim naming a player the graph has no record of is unverified because absence is not refutation - and advertises that distinction as a contribution. Study II statistics are now recomputed from the raw per-claim records. Coverage is unaffected at 0.495: the gate withholds both.
2	Rule-only refutations: twenty-one corrected to **19**.	19 records carry a rule-failure entry, one each. The figure 21 matches nothing in the artifacts.
3	Extraction coverage 96.9% corrected to **96.8%**.	1,056/1,091 = 96.79%.

4	Status breakdown rewritten.	It listed the 35 unparseable sentences inside the 1,056 parsed ones, so the breakdown did not sum to its own subject.
5	Commonest refutation corrected.	Relocation to the wrong tournament accounts for 98 of the refutations. The largest group, 152, is a goal event not on record at the stated minute. Wrong-tournament is the next largest and is now described as such.
6	Contribution 5 no longer claims a scale measurement.	It advertised a measurement that was cut from the results for running across two inference stacks, and which the limitations section lists as an open question. The paper claimed and retracted the same result. The contribution is now the error analysis alone; the count stays at six.

6.3 Ablation rows that read as refutations

Three of the seven ablation rows were reported without the column that explains them, and two read as refutations of the paper's own claims. Every figure needed was already in the stored report.

Row	Looks like	Resolution
Kind-agreement check removed	Accuracy **0.804**, above the full system's 0.796 - a check that hurts.	The paired test gives p = 0.625 on 4 discordant cases: noise. What the check buys is 0.012 hallucination for 0.019 coverage. Now stated in Section VIII.
Asker's statements override records	Hallucination **0.094**, well below 0.149 - trusting the asker looks safer.	Coverage falls 0.943 to 0.748 and accuracy drops significantly (p = 1.49e-08). The configuration abstains its way to a better rate - the exact failure the metrics section warns about, demonstrated inside the paper's own table. Now stated.
Numeric grounding check removed	No change at all (p = 1) - a check that does nothing.	The same reason the verification gate shows no decision-level effect: both police prose, and the default configuration writes none. The gate's no-op had a subsection; this one had no mention. Now stated.

Study II's discarded fragments are now quantified too. The paper said 300 trailing fragments were discarded as a sampling artifact without stating the share: it is 21.6% of generated sentences, but **0.89 per prompt against a fixed 80-token budget**, which is the evidence that it is one severed tail per generation rather than a selection among claims.

6.4 Cross-references, statistics and format

#	Correction	Evidence
7	Broken cross-reference fixed.	The conclusion cited "the capability effect reported in Section VIII". That section had been removed; Section VIII is Ablation Studies, so the pointer resolved to the wrong content. It now points to Section XII, where the finding survives as a limitation.
8	References is no longer a numbered section.	Heading 1 auto-numbers in the IEEE template, so References printed as a numbered section. It now takes the style the template itself uses for References and Acknowledgment. Sections are I-XIII with References unnumbered.
9	The statistical test is now defined.	The paper reported p-values without stating the test, so they could not be checked. It now states that concordant pairs are discarded and the discordant split tested two-sided against Binomial(n, 1/2), using the exact tail.
10	Verification-gate paragraph rewritten.	The earlier wording could be read as contradicting the controlled experiment that follows it. It now names both measurements and says which is silent about the gate by construction.
11	Unsupported-figure wording corrected to "from four to six".	An earlier draft implied the gated arm carried none. The counts are 4 and 6.

12	Three wrong author names corrected in the bibliography.	Verified against ACL Anthology and the ICLR listing. None reached the PDF, which cites all three with "et al.", but the bibliography attributed work to researchers who did not write it.

6.5 Abstract and figures

#	Correction	Evidence
13	The abstract named one unseen-world figure where two were needed. It now reports **accuracy 0.835 and selective accuracy 0.909** separately.	It quoted 0.945 accuracy and 1.000 selective accuracy and then said "which falls to 0.909". The nearest antecedent was correct, but a reader could take it as 0.945 falling to 0.909. Confirmatory accuracy is 0.835.
14	**Fig. 1 redrawn** as a vertical single-column diagram sized for the page.	Its labels were specified at 6.2 pt inside a 7.6 in figure and the figure was placed at about 3.3 in, so they printed at roughly **2.5 pt** - in the file, but not readable in the paper. Every font size is now the size it prints at; the smallest label is 6.0 pt.
15	`--audit` mode added to the figure script.	It reports the smallest printed label, checks every label against the width of its own box, and fails if the width the builder places the figure at differs from the width it was drawn for - the mismatch that caused the defect.
16	Fig. 1's caption distinguishes the three-valued rule engine from the four verification statuses, and carries the shading key.	They are separate layers. The in-figure legend was removed in favour of the caption, which prints at caption size.
17	SVG emitted beside every figure PNG.	Word rasterises embedded images on export: measured 200, 200 and 226 dpi for Figs. 1-3 in the delivered PDF. The vector sources let the camera-ready substitute scalable art.

6.6 Implementation defects found while auditing

#	Defect	Consequence
13	The ablation switch reached one of three guard layers.	Two successive runs of the gate experiment produced byte-identical arms. The experiment was unmeasurable until the switch reached all three.
14	The configuration literal omitted the local-model provider.	The documented local-model path raised a validation error on import, so those baselines could not be run at all.
15	Parser fallback was not recorded per case.	36% of parser runs fall back to the embedding interpreter. Visible only in stderr, so the score could be read as the model's understanding when a third of it was not.

A reporting failure on the audit's own side is recorded for completeness: a page count was once read from a stale PDF because build output had been suppressed and a syntax error passed unnoticed. Build output is no longer suppressed.

7. Unresolved items

These are stated in the paper's limitations section and repeated here so that what the evidence does not cover is legible in one place. The first two are the ones a reviewer should weigh most heavily.

Item	Status	What would settle it
Question relevance	Not implemented	A relevance gate between the resolved goal and the question, independent of the evidence gate. Until then the system answers out-of-scope questions confidently: on the fifteen questions no policy clause covers it abstained in none. Understanding maps an out-of-scope question onto the nearest in-scope goal and the system answers that question correctly, citing real clauses. Verification establishes that a claim follows from evidence, not that it answers the question asked.
Verifier precision and recall	Unmeasured	Independent ground-truth labels, authored without reference to the rule base. The verifier is correct by construction with respect to its own graph, so a graph error would propagate silently.
Independently authored evaluation	Not done	A second person writing questions and gold answers without seeing the rule base. The rule base, world generators and oracle were written by one person.
Model capability effect	Open question	One inference stack serving both model sizes on the same split. The 1.5B and 7B runs used different stacks, so the comparison is not controlled.
Prose-faithfulness metric	Not implemented	Needed to quantify the gate's contribution beyond counting changed answers and unsupported figures.
Run-to-run variance	Estimated from two runs	Three to five seeded repeats, giving a variance estimate rather than a two-sample difference.
Conference reference pagination	Partly verified	Pages and DOIs confirmed for five journal and five conference entries. Three entries publish through OpenReview without page numbers; one NeurIPS entry's volume is confirmed but its page range is not.

8. Artifact inventory

Every figure in the paper traces to one of these files. Per-system record hashes are listed in docs/EXPERIMENT_MANIFEST.json.

Path	Contents	Size
docs/paper/PRAXIS_IEEE.pdf	the submission	468 KB
docs/paper/PRAXIS_IEEE.docx	editable source	315 KB
eval_reports/research/paper/	test split, per-question records	14 record files
eval_reports/research/confirm/	confirmatory split records	5 record files
eval_reports/research/gate-ablation-v3/	gate experiment, both arms	2 record files
artifacts/football/results/claims.jsonl	Study II, per-claim records	675 KB
docs/EXPERIMENT_MANIFEST.json	record hashes and system inventory	12 KB
docs/audit/CHANGE_LOG.md	every correction, with evidence	15 KB
docs/audit/STATISTICAL_VALIDATION.md	all paired tests recomputed	3 KB
docs/audit/GATE_EVIDENCE.md	gate claims against code and records	2 KB
docs/audit/UNRESOLVED.md	open items	1 KB

9. Statement

The experiments in this paper were run on generated enterprise worlds and a public record of World Cup goals. No human subjects, personal data or proprietary corpora were involved. The language model used as a baseline and as a semantic parser is openly available; the model in Study II was trained from random

initialization on the corpus described, and its parameter provenance is asserted programmatically rather than claimed.

Every number in the paper is recomputed from a stored record by scripts/audit_metrics.py and scripts/verify_claims.py, both of which exit non-zero on any mismatch. The results reported above are the output of those scripts at the commit named on the first page, not a transcription of an earlier run.